When to use?

Harmonic mean is used to calculate the average value when the values are expressed as value/unit. Since the speed is expressed as km/hour, harmonic mean is used for the calculation of average speed.

Relationship among the averages:

In any distribution when the original items are different the A.M., G.M. and H.M would also differ and will be in the following order:

$$\textbf{A.M.} \geq \textbf{G.M} \geq \textbf{H.M}$$

5.3.4 Median

Median is the value of the variable which divides the whole set of data into two equal parts. It is the value such that in a set of observations, 50% observations are above and 50% observations are below it. Hence the median is a positional average.

(a) Median for Ungrouped or Raw data:

In this case, the data is arranged in either ascending or descending order of magnitude.

(i) If the number of observations n is an odd number, then the median is represented by the numerical value of x, corresponds to the positioning point of $\frac{n+1}{2}$ in ordered observations. That is,

$$\text{Median} = \text{value of } \left(\frac{n+1}{2}\right)^{th} \text{ observation in the data array}$$

(ii) If the number of observations n is an even number, then the median is defined as the arithmetic mean of the middle values in the array That is,

$$\text{Median} = \frac{\textit{value of } \left(\frac{n}{2}\right)^{th} \textit{ observation} + \textit{value of } \left(\frac{n}{2}+1\right)^{th} \textit{ observation}}{2}$$

Example 5.14

The number of rooms in the seven five stars hotel in Chennai city is 71, 30, 61, 59, 31, 40 and 29. Find the median number of rooms

Solution:

Arrange the data in ascending order 29, 30, 31, 40, 59, 61, 71

$n = 7$ (odd)

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